LAB 7: FUZZ TESTING

Input validation hardening with Hypothesis

Repository and handout mapping

The handout calls the program `processor.py` and the tests `test_processor.py`. In the assigned repository, the matching Lab 7 files are `self_storage.py` and `test_self_storage.py`. This submission uses the repository filenames.

Handout name	Repository name
processor.py	self_storage.py
test_processor.py	test_self_storage.py

Changes and results

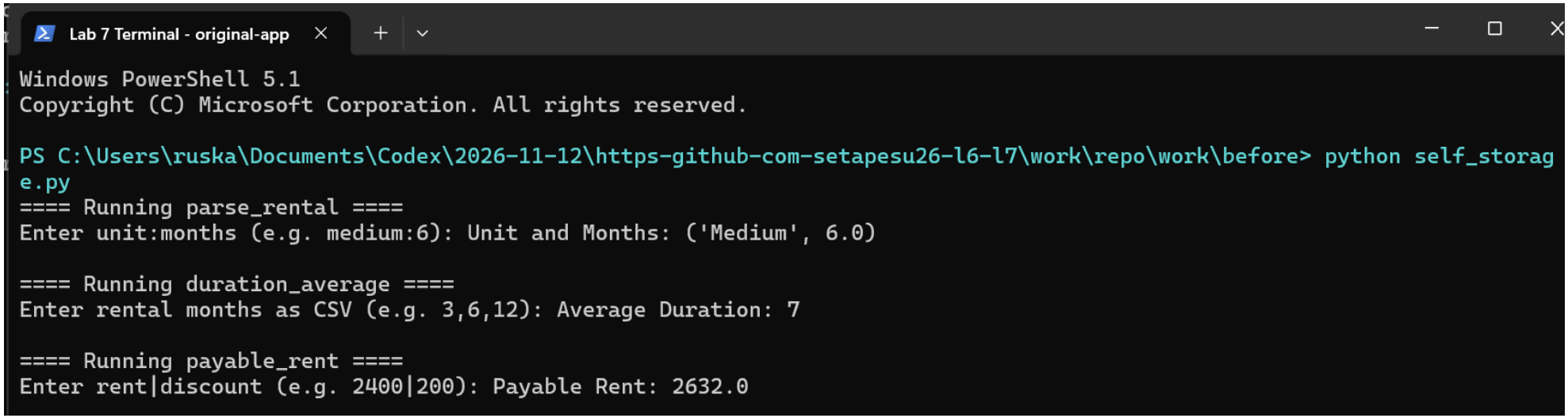
The application validates input types, delimiters, empty fields, numeric formats, finite values, and non-negative amounts. Invalid input raises a clear `ValueError`. The interactive program catches validation errors and continues. Rent calculation retains the existing 18% tax, discount, and zero-floor behavior; duration average retains integer floor division.

Hypothesis minimized the original failures to `None` for each function. The updated application was then tested with Hypothesis: all three property tests passed. The evidence pages contain the actual terminal screenshots provided for the original run, the failing fuzz run, the fixed app run, and the passing tests.

Run from the Lab 7 directory

```
python -m pip install hypothesis pytest
python self_storage.py
python -m pytest -q test_self_storage.py
```

1. Original application run



2. Hypothesis failures before the fix

```
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before> python -m pytest -q --tb=line t
est_self_storage.py
FFF                                                                      [100%]
================================================== FAILURES ==================================================
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_parse_rental_is_robust(
        value=None,
    )
C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before\self_storage.py:2: AttributeError:
'NoneType' object has no attribute 'split'
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_duration_average_is_robust(
        value=None,
    )
C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before\self_storage.py:6: AttributeError:
'NoneType' object has no attribute 'split'
E   AttributeError: 'NoneType' object has no attribute 'split'
    Failing test case: test_payable_rent_is_robust(
        value=None,
    )
C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before\self_storage.py:10: AttributeError:
 'NoneType' object has no attribute 'split'
============================================ short test summary info =============================================
FAILED test_self_storage.py::test_parse_rental_is_robust - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_self_storage.py::test_duration_average_is_robust - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_self_storage.py::test_payable_rent_is_robust - AttributeError: 'NoneType' object has no attribute 'split'
3 failed in 1.04s
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before>
```

3. Fixed application run

```
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\work\before> Set-Location -LiteralPath '..\..\Lab 7'
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7> Get-Location

Path
----
C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7


PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0

PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7>
```

4. Hypothesis tests after the fix

```
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7> python -m pytest -q test_self_storage
.py
...                                                                                                                    [100%]
3 passed in 3.83s
PS C:\Users\ruska\Documents\Codex\2026-11-12\https-github-com-setapesu26-l6-l7\work\repo\Lab 7>
```